Understated Haptics: Physicalization of Tactile Memories through a Co-creative Practice between Human and Nonhuman

Ivana Chaloska*
Keio University Graduate School of Media Design
Japan
ichaloska@kmd.keio.ac.jp

Tatsuya Saito
Keio University Graduate School of Media Design
Japan
tatsuyas@kmd.keio.ac.jp

Junichi Yamaoka
Keio University Graduate School of Media Design
Japan
yamaoka@kmd.keio.ac.jp

Kouta Minamizawa
Keio University Graduate School of Media Design
Japan
kouta@kmd.keio.ac.jp

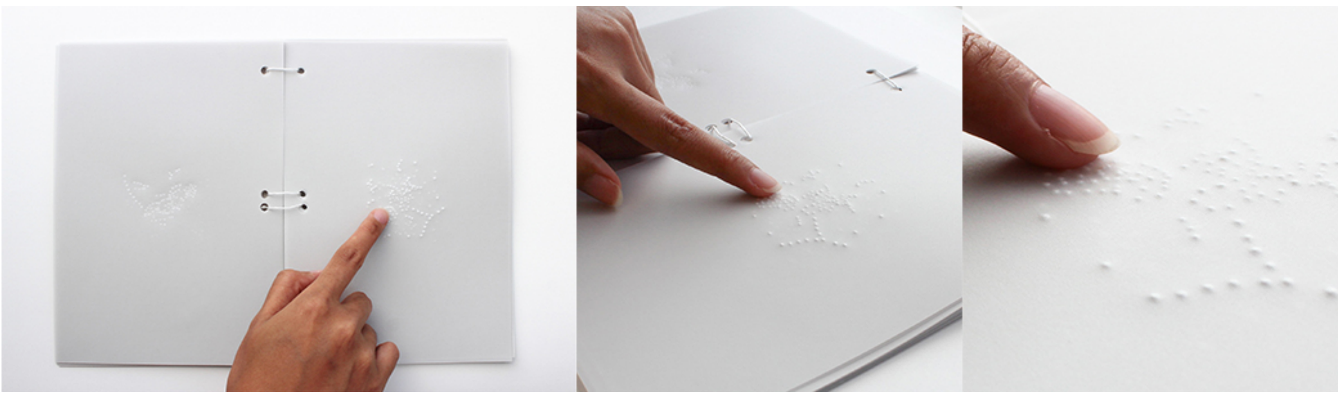

Figure 1: Understated Haptics is a collection of tactile co-memories from humans and nonhumans who had collaborated before.

Abstract

Figuratively speaking, when two surfaces, such as human skin and an object's outsideness, deliberately interact - a third, collateral, and temporary surface is created. This work presents an artistic framing of tactility as a third surface: a tactile co-memory generated through tactile interactions in a Co-creative Practice between humans and nonhumans. In this practice, human and nonhuman tactility is documented through drawings and imprints, ultimately translated into haptic expressions. Understated Haptics demonstrates an artistic intersection where haptic artifacts and more-than-human prospects are critically considered in anticipated creative processes.

CCS Concepts

• **Human-centered computing → Haptic devices**; **Visualization theory, concepts and paradigms**; **Information visualization**; • **Applied computing → Media arts**.

SA Art Papers '24, December 03–06, 2024, Tokyo, Japan

ACM ISBN 979-8-4007-1133-6/24/12
https://doi.org/10.1145/3680530.3695458

Keywords

tactility, tactile interactions, more-than-human, co-creation

ACM Reference Format:
Ivana Chaloska, Tatsuya Saito, Junichi Yamaoka, and Kouta Minamizawa. 2024. Understated Haptics: Physicalization of Tactile Memories through a Co-creative Practice between Human and Nonhuman. In *SIGGRAPH Asia 2024 Art Papers (SA Art Papers '24), December 03–06, 2024, Tokyo, Japan.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3680530.3695458

1 Introduction

Understated Haptics refers to the analog physicalization of tactile memories generated in a Co-creative Practice between humans and nonhumans. In this context, the nonhumans are inanimate objects collected from unanticipated walks in nature, including stones, sea shells, tree branches, and pine cones.

Figuratively speaking, when two surfaces, such as human skin and an object's outsideness, deliberately interact - a third, collateral, and temporary surface is created (Figure 2). Although physically imperceptible, we can indulge in its interpretation, leveraging our imagination and aesthetic sensibility. Understated Haptics is an artifact that embodies the interpretations of tactility as a third surface, a byproduct of tactile interactions utilizing graphic materials produced in a Co-creative Practice in which humans and nonhumans collaborate to draw tactile sensations and imprint tactile mediums. These graphic materials are artistically integrated into a cohesive composition, which is then translated into a haptic drawing - a physical metaphor of shared memory, *a tactile co-memory*. Understated Haptics is a collection of tactile co-memories from humans and nonhumans who had collaborated before (Figure 1).

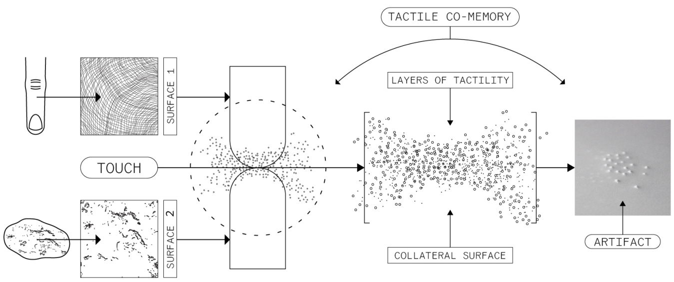

Figure 2: The diagram illustrates tactility as a figurative collateral surface between two interacting surfaces: the human skin and the object's outsideness. Different layers of tactility are activated in a tactile interaction, composing a tactile co-memory embodied with a haptic drawing.

To contribute to more-than-human perspectives in creative research, we designed the Co-creative Practice through experimental tactile activities that enable us to establish creativity that is not confined to humans but is a spectrum of performances by humans and nonhumans [Wakkary 2021], therefore, in our work we *create with*, rather than control of.

When we *tactually* encounter another existence, a unique exchange occurs as we partially lend our association and emotional response to it, while in return, this existence allures and liberates our perception. In this regard, among the broad ontological and materialist points of view such as *'object-oriented ontology'* [Harman 2018] and *'vital materiality'* [Bennett 2010], this work disputes the notion that our nonhuman collaborators - despite being inanimate objects - are situated in passiveness, conceptualizing their agency as actively influencing the human. Under this framework, tactility is not a physical margin but a dynamic medium for imagination and co-creation.

Building on this foundation, Understated Haptics demonstrates an artistic intersection where haptic artifacts and more-than-human prospects are critically considered in anticipated creative practices.

2 Related Works

Tactility is a *quality* that emerges when our perception is alerted by the sense of touch. Tactility helps us recognize and experience matter through touch, beginning from our body and extending from its spatial position to the substance representing any other existence. The initial motivation for Understated Haptics is anchored in foundational concepts of touch, tactility, and haptics, delving into philosophical and artistic territory [Aristotle 1984; Gibson 1962; Katz 2013; Marks 2000; Pallasmaa 2005].

Of significant importance to this work are various tactile experiments and artifacts designed for a deliberate and mindful tactile exploration by the surrealist artist Jan Švankmajer [Švankmajer 2014]. Švankmajer aimed to prove that tactile memory and imagination exist, believing that art's purpose today is to free us from reality's constraints and that touch could play a significant role in this liberation [Driscoll 2020]. In correlation, Rosalyn Driscoll refers to philosopher Richard Kearney's suggestion that the next stage of imagination will be more inclusive of others, and the concept of touch, with its mutual and reciprocal nature, could serve as a model for this new way of imagining [Driscoll 2020].

More-than-human perspectives in art, design, and HCI also play a pivotal role in our work. These perspectives challenge human-centered thinking, resist human/nonhuman dichotomy, and investigate the parameters of nonhuman agency, pluriversality, and co-creation [Akama et al. 2020; Coskun et al. 2022].

For example, sculptor Martin Kuhn immersed in a sculpting act bound to listening to a rock, to establish nonhuman agency reliant on human eagerness to engage [Fredriksen and Haukeland 2023]. Other artists, intended to create a collaborative novel with an oak tree [Anderson 2014]; or let rain and waves inscribe themselves on sensitized paper, returning creativity to nature [Närhinen 2022].

In *thing ethnography*, several household objects, equipped with sensors, explored human activities from their perspective, revealing the captivating potential of nonhumans to act as co-ethnographers [Giaccardi et al. 2016]. Related to this are concepts like *agential realism* [Barad 2007] and *tool-being* [Harman 2002], which Charlotte Nordmoen has probed, similarly, investigating how sensors can act as co-investigators and co-creators in shaping knowledge [Nordmoen and McPherson 2023].

Drawing from these works, we affirm that envisioning co-creation with the nonhuman is impossible without stepping out of the ordinary and embracing abstraction, imagination, and experimentation. However, we diverge by focusing explicitly on co-creation mediated through tactility, a dimension yet to be disseminated in more-than-human explorations.

Integrating these theoretical and pragmatic directions has facilitated the Understated Haptics artifact, guided by two foundational criteria: 1) Co-creative Practice between human and nonhuman, and 2) Tactility as a mediator for co-creation.

3 Co-creative Practice between Humans and Nonhumans

Figure 3: The nonhuman participants in the Co-creative Practice: stones, sea shells, tree branches, and pine cones.

The Understated Haptics artifact was created based on graphic materials generated through a Co-creative Practice between humans and nonhumans. We designed this practice to stimulate the human awareness of nonhumans and their impact in an act of co-creation, thus challenging anthropocentrism. The nonhuman

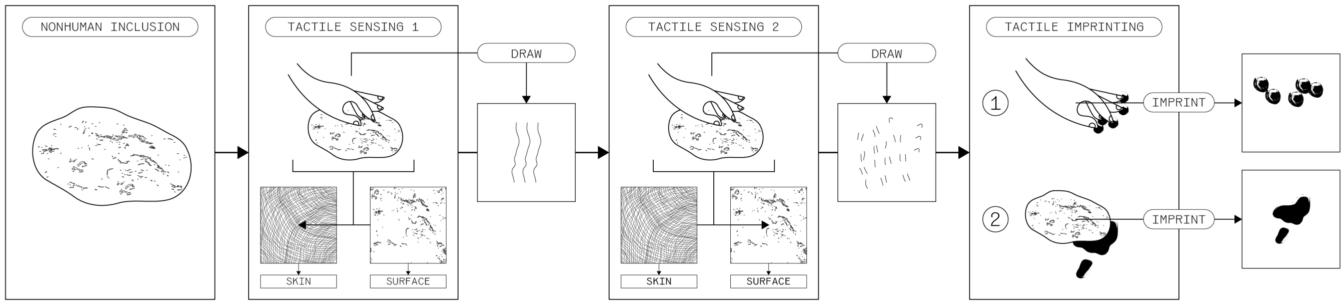

Figure 4: The diagram illustrates the activity flow in the Co-creative Practice between humans and nonhumans. It begins with the inclusion of the nonhuman as a collaborator and continues to: Tactile Sensing 1 (exploring and drawing human tactile sensation in tactile interaction with the nonhuman), Tactile Sensing 2 (imagining and drawing nonhuman tactile sensation in tactile interaction with the nonhuman), and Tactile Imprinting 1 and 2 (imprinting of human and nonhuman tactile medium).

collaborators in this practice are stones, sea shells, tree branches, and pine cones (Figure 3).

The Co-creative Practice titled *'Merge: Tactile Co-memories'* consists of four creative activities: Tactile Sensing 1, Tactile Sensing 2, Tactile Imprinting 1, and Tactile Imprinting 2 (Figure 4).

Tactile Sensing 1 is an activity where human and nonhuman participants engage in five tactile interactions: tap, press, hold, caress, and trace. Each tactile interaction is performed with suspended vision, occurring at 3-5-minute intervals.

After each interaction, the human participants created a single drawing using black ink pens and paper to express their interpretation of the tactile sensations. This activity represents conventional tactile sensing, as it refers to predetermined human knowledge, answering the question: '*What are you sensing when you touch the nonhuman (object)?*'.

Tactile Sensing 2 is the subsequent activity where the human and nonhuman participants repeat the five tactile interactions, again with a suspended vision, during separate time intervals. The critical difference in this activity is that the humans were coerced to *perspective-taking* [Davis et al. 1996] (attempting to sense from the nonhuman point of 'tactility').

After each interaction, the human participants created a single drawing using black ink pens and paper to express their interpretation of the nonhuman tactile sensations. This activity represents alternative tactile sensing, an imaginative act based on human illiteracy. It addresses the question: '*What is the nonhuman (object) sensing when you touch it?*'.

Tactile Imprinting 1 and 2 are the concluding creative activities in which humans and nonhumans collaborate to produce additional graphic evidence of tactility. Using black ink pads, humans imprinted their tactile medium (fingers' skin) on paper while performing the five tactile interactions. Subsequently, using the same technique, the humans assisted the nonhumans in imprinting their tactile medium (objects' surface).

To implement and validate the Co-creative Practice, we organized an interactive workshop with seven human and seven nonhuman participants. For each creative activity, the human participants received step-by-step guidelines and paper sheets for tactile sensing and imprinting corresponding to each tactile interaction: tap, press, hold, caress, and trace. The creative activities were followed by a discussion and a reflection session. The discussion session was guided by open-ended questions divided into four topics: general experience, tactile perception, tactile interaction, and nonhuman role. Data was collected using a mobile phone audio recording and later transcribed for a more accessible assessment. In the reflection session, data was gathered using a questionnaire with open-ended and close-ended questions.

The complete data from the workshop was divided into two modules: 1) audio transcripts and questionnaire responses, and 2) drawings of tactile sensations and imprints of tactile mediums. The first data module, *audio transcripts and questionnaire responses*, was analyzed qualitatively using a thematic analysis based on the six-phase framework by Braun and Clarke [Braun and Clarke 2021]. The findings suggested that engaging in tactile interactions with the nonhumans facilitated a transition from perceiving them as passive objects to recognizing them as collaborators. This led to humans questioning their individual human biases, including the concept of *consent* for nonhuman participation.

4 Understated Haptics Artifact

The second module of data from the workshop, *drawings of tactile sensations and imprints of tactile mediums*, was subdivided into four modules of graphic materials, each representing a separate visual evidence of tactility: 1) a drawing of human tactile sensation, 2) a drawing of nonhuman tactile sensation, 3) an imprint of human tactile medium (skin), and 4) an imprint of nonhuman tactile medium (object's surface). Seven collaborative couples of human and nonhuman participants produced the four graphic materials for each of the five tactile interactions: tap, press, hold, caress, and trace. This visual data was utilized in an artistic process, resulting in the Understated Haptics artifact.

In the first step of the artistic process, each group of four graphic materials corresponding to a tactile interaction was artistically manipulated with tracing and stippling techniques. Tracing includes creating a handmade copy of a graphic material by drawing over it using thin and translucent paper, while stippling is an art technique that creates drawings with numerous dots. The graphic materials were translated into simple dot drawings using tracing paper sheets and a black ink pen.

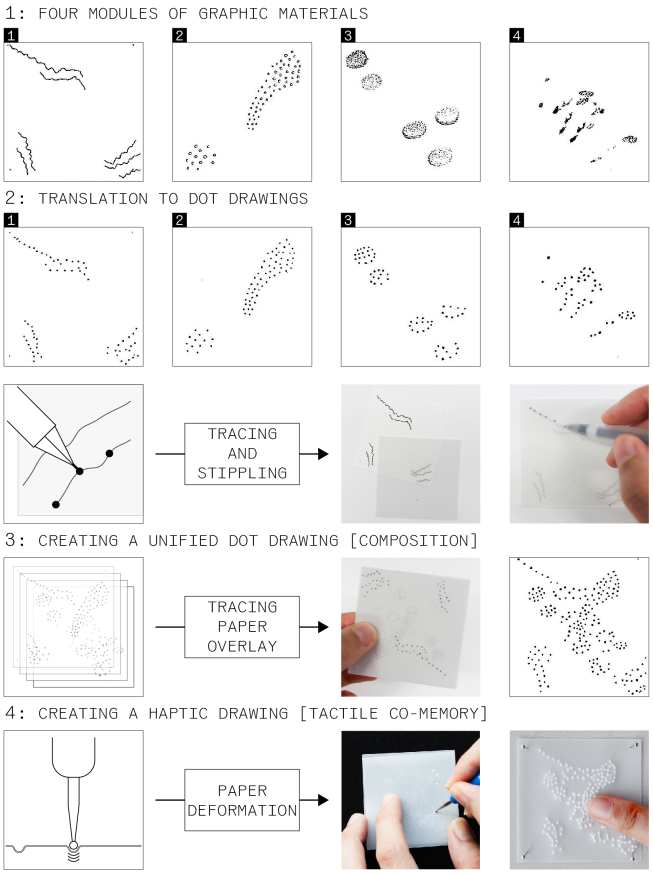

Figure 5: The artistic process of the Understated Haptics artifact: 1) 4 modules of graphic material: 1. drawing of human tactile sensation, 2. drawing of nonhuman tactile sensation, 3. imprint of human tactile medium, 4. imprint of nonhuman tactile medium; 2) translation to dot drawings; 3) creating a unified dot drawing; and 4) creating a haptic drawing.

The next step of the artistic process involved tracing paper overlay, a technique commonly used in art, design, and architecture in which multiple tracing papers are layered over each other. The tracing paper sheets with the dot drawings were overlaid, culminating in unified dot drawings (compositions) corresponding to each collaborative couple's five tactile interactions. These compositions represent visual concepts of tactility as a third surface between the two interacting surfaces.

In the final step of the artistic process, with paper deformation, the unified compositions were translated into haptic drawings: *a metaphorical embodiment of tactile co-memories*. To create a haptic drawing, a tracing paper sheet with a unified composition was placed over an aligned, empty paper sheet on a felt fabric, which was itself laid on a hard surface. Using an embossing stylus, each drawn dot of the composition was gently pressed to create a raised dot without piercing the empty paper sheet, producing a texture of haptic dots resembling Braille.

This artistic process resulted in 35 distinct haptic drawings: seven haptic drawings from seven collaborative couples for each of the five tactile interactions: tap, press, hold, caress, and trace. The haptic drawings were compiled in a book forming the Understated Haptics artifact: *'a collection of tactile co-memories from humans and nonhumans who had collaborated before'* (See Figure 1).

Figure 5 depicts the artistic process of the Understated Haptics artifact.

5 Exhibiting Understated Haptics

The Understated Haptics artifact and several components demonstrating representative examples of every step of its artistic process were exhibited at a group exhibition.

Figure 6: Understated Haptics' exhibition display and audience participation in the Co-creative Practice.

The Understated Haptics artifact was displayed with clear instructions to allow the audience to interact with it, facilitating a tactile exploration of each haptic drawing. The audience was also invited to participate in a simplified format of the Co-creative Practice, explicitly focusing on shared tactile sensations with nonhuman collaborators (stones, sea shells, tree branches, and pine cones). Figure 6 shows the exhibition display of the Understated Haptics artifact and how the audience participates in the Co-creative Practice.

The audience reflected on their experience, tactile perception, and the nonhuman role through a brief questionnaire with open-ended questions. 20 gathered responses were qualitatively assessed using a thematic analysis [Braun and Clarke 2021], providing insights on heightened tactile sensitivity, perceptual merging of human and nonhuman matter, and altered awareness of nonhuman existence. Public participation in the Co-creative Practice has reaffirmed that its activities deepen the human perception of nonhumans as collaborators, challenging the anthropocentric thought of nonhumans as passive objects, tools, or resources.

6 Discussion

The Co-creative Practice and the Understated Haptics artifact, propose a concept for co-creation with the nonhuman that aspires to blur distinctions between subject/object, active/passive, and human/nonhuman through tactile interactions. We use the language of touch—surface to surface, *"body on body"* [Jensen 2004]—to speculate *nonhuman tactility*, as in the human truth we can only imagine it. *But do we, humans, even experience tactility beyond our body if we don't touch the 'otherness'?*

Consequently, we can assume that our stones, seashells, tree branches, and pine cones, through their tactile medium (surface), perform, stimulate, guide, and configure tactility, influencing the creation of tactile co-memories in ways both perceived and concealed [Harman 2018]. In the context of Understated Haptics, we contend that the nonhumans are the shimmering essence in shaping tactility. Hence, their agency is inherent, not something we need to bestow.

A significant limitation of this work is our fundamental human bias and the lack of established frameworks and methods for co-creation with nonhumans, resulting in continuous experimentation and subjectivity. Moreover, the Understated Haptics artifact relied entirely on analog processes and individual aesthetic judgment, despite the potential for a more mechanical approach.

This work is distinctive in its utilization of drawings and imprints to capture and represent human and nonhuman tactility. During the exhibition, the audience produced 116 drawings of human and nonhuman tactile sensations (Figure 7), which have yet to be processed. There is a feasible prospect of leveraging these drawings for data physicalization [Jansen et al. 2015], incorporating technology as an additional layer of nonhuman participation.

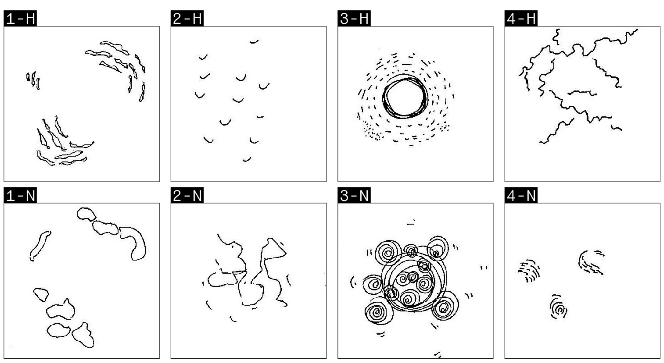

Figure 7: Samples from the 116 drawings collected during the Understated Haptics' exhibition. The top row refers to drawings of human tactile sensation (1-H, 2-H, 3-H, and 4-H - human participants), and the bottom row refers to the non-human tactile sensation (1-N, 2-N, 3-N, and 4-N - nonhuman collaborators).

7 Conclusion

Our experimentation with tactile interactions frames tactility as a viable mediator between humans and nonhumans. The artistic translation of drawings and imprints into a haptic artifact became a crafted metaphor for a third surface between two interacting surfaces: liminality where human and nonhuman tactility is preserved and revisited. The Co-creative Practice has demonstrated that a peculiar empathy is evoked through curated imagination, challenging superiority dynamics, and conceptualizing nonhuman agency. In summary, Understated Haptics nurtures a modest contribution to more-than-human perspectives in artistic and design research.

As a closing remark, ***understated*** *in* ***haptics*** *refers to dormant, barely perceptible, subtle, restrained, uncanny, and undetermined sensations. It holds a captivating simplicity, drawing into a pluriverse where even the most minor and overlooked touches carry significance. Understated Haptics contemplates a delicate play between perception and interpretation, depicting tactility that is not obvious or impressive yet fragile and complex.*

Acknowledgments

We thank all our participants for their significant input and the reviewers for their constructive feedback.

References

Yoko Akama, Ann Light, and Takahito Kamihira. 2020. Expanding Participation to Design with More-Than-Human Concerns. In *Proceedings of the 16th Participatory Design Conference 2020 - Participation(s) Otherwise - Vol 1*. ACM, New York, NY, USA, 11. https://doi.org/10.1145/3385010.3385016

Kayla Anderson. 2014. Object intermediaries: how new media artists translate the language of things. In *ACM SIGGRAPH 2014 Art Gallery* (Vancouver, Canada) *(SIGGRAPH '14)*. Association for Computing Machinery, New York, NY, USA, 352–359. https://doi.org/10.1145/2601080.2677712

Aristotle. 1984. On the Soul. In *The Complete Works: The Rev. Oxford Translation*, Jonathan Barnes (Ed.). Princeton University Press, 641–692.

Karen Barad. 2007. *Meeting the Universe Halfway: Quantum Physics and the Entanglement of Matter and Meaning.* Duke University Press.

Jane Bennett. 2010. *Vibrant Matter: A Political Ecology of Things.* Duke University Press.

Virginia Braun and Victoria Clarke. 2021. *Thematic Analysis: A Practical Guide.* SAGE.

Aykut Coskun, Nazli Cila, Iohanna Nicenboim, Christopher Frauenberger, Ron Wakkary, Marc Hassenzahl, Clara Mancini, Elisa Giaccardi, and Laura Forlano. 2022. More-than-human Concepts, Methodologies, and Practices in HCI. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) *(CHI EA '22)*. Association for Computing Machinery, New York, NY, USA, Article 150, 5 pages. https://doi.org/10.1145/3491101.3516503

Mark H. Davis, Laura Conklin, Ashley Smith, and Carole Luce. 1996. Effect of perspective taking on the cognitive representation of persons: A merging of self and other. *Journal of Personality and Social Psychology* 70, 4 (1996), 713–726. https://doi.org/10.1037//0022-3514.70.4.713

Rosalyn Driscoll. 2020. Rosalyn Driscoll: Sensory Imagination. *Tangible Territory Journal* 1, 1 (Nov. 2020). https://tangibleterritory.art/journal/issue1/rosalyn-driscoll-sensory-imagination/

Biljana C. Fredriksen and Per Ingvar Haukeland (Eds.). 2023. *Crafting Relationships with Nature through Creative Practices.* Scandinavian University Press.

Elisa Giaccardi, Nazli Cila, Chris Speed, and Melissa Caldwell. 2016. Thing Ethnography: Doing Design Research with Non-Humans *(DIS '16)*. Association for Computing Machinery, New York, NY, USA, 377–387. https://doi.org/10.1145/2901790.2901905

James J. Gibson. 1962. Observations on Active Touch. *Psychological Review* 69, 6 (1962), 477–491. https://doi.org/10.1037/h0046962

Graham Harman. 2002. *Tool-Being: Heidegger and the Metaphysics of Objects.* Open Court.

Graham Harman. 2018. *Object-Oriented Ontology: A New Theory of Everything.* Penguin UK.

Yvonne Jansen, Pierre Dragicevic, Petra Isenberg, Jason Alexander, Abhijit Karnik, Johan Kildal, Sriram Subramanian, and Kasper Hornbæk. 2015. Opportunities and Challenges for Data Physicalization. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (New York, NY, USA, 2015-04-18) *(CHI '15)*. Association for Computing Machinery, 3227–3236. https://doi.org/10.1145/2702123.2702180

Derrick Jensen. 2004. *A Language Older Than Words.* Chelsea Green Publishing.

David Katz. 2013. *The World of Touch.* Psychology Press.

Laura U. Marks. 2000. *The Skin of the Film: Intercultural Cinema, Embodiment, and the Senses.* Duke University Press.

Charlotte Nordmoen and Andrew P. McPherson. 2023. What the Sensor Knows: More-Than-Human Knowledge Co-Production in Wood Carving. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference* (Pittsburgh, PA, USA) *(DIS '23)*. Association for Computing Machinery, New York, NY, USA, 779–789. https://doi.org/10.1145/3563657.3596075

Tuula Närhinen. 2022. Do It Yourself, Rain! Dabbling Drops, Splashes, and Waves: Experiments in Art and Science. *Leonardo* 55, 6 (2022), 627–634. https://doi.org/10.1162/leon_a_02280

Juhani Pallasmaa. 2005. *The Eyes of the Skin: Architecture and the Senses.* Wiley.

Jan Švankmajer. 2014. *Touching and Imagining: An Introduction to Tactile Art.* Bloomsbury Publishing.

Ron Wakkary. 2021. *Things We Could Design: For More Than Human-Centered Worlds.* MIT Press, Cambridge, MA.